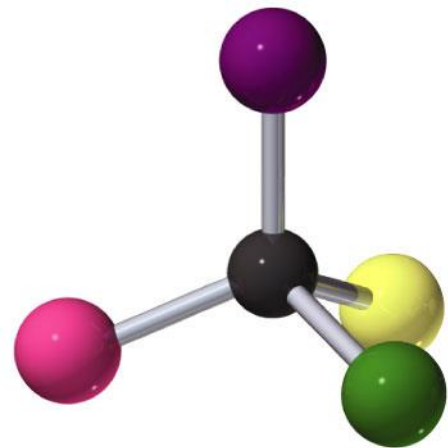
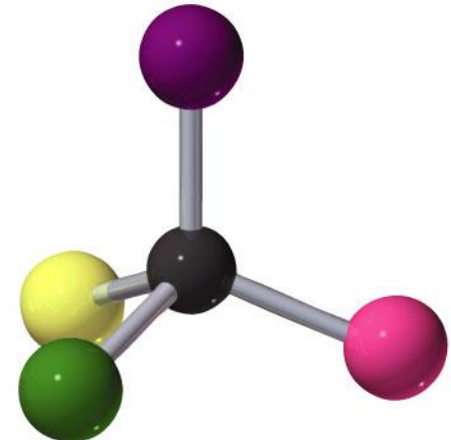




© 2011 Pearson Education, Inc.

Chapter 4



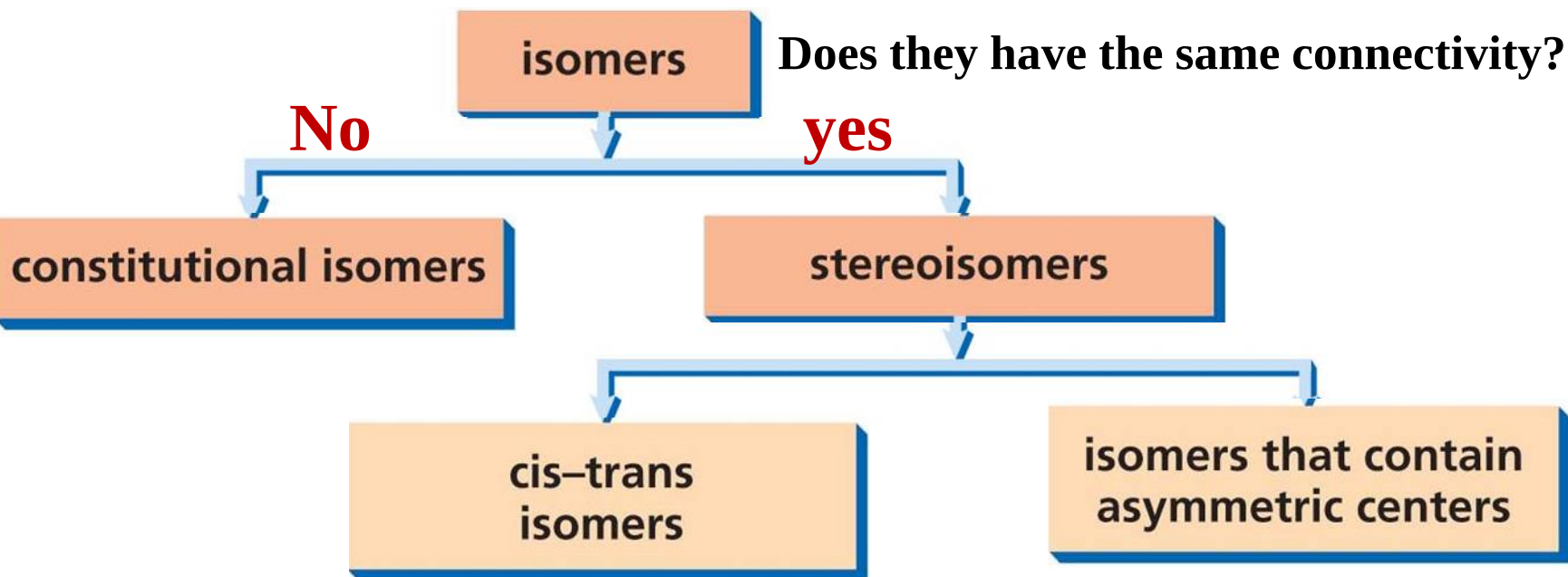
© 2011 Pearson Education, Inc.

STEREOCHEMISTRY

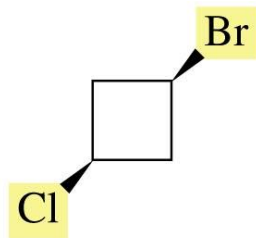
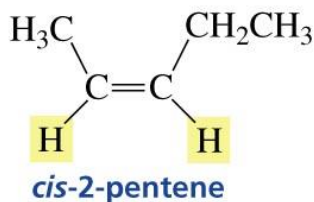
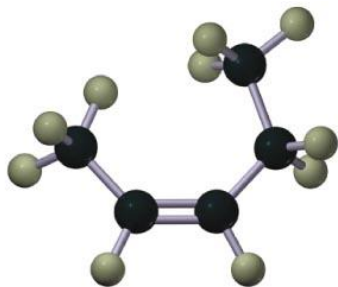
**THE ARRANGEMENT OF ATOMS IN SPACE;
THE STEREOCHEMISTRY OF ADDITION REACTIONS**

Isomers

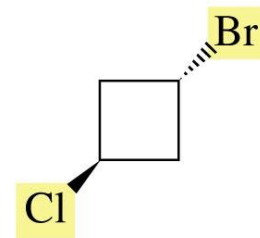
Non-identical compounds having the same molecular formula



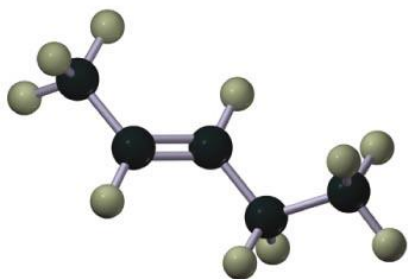
Cis-Trans Isomers in Alkenes and Rings



cis-1-bromo-3-chlorocyclobutane



trans-1-bromo-3-chlorocyclobutane

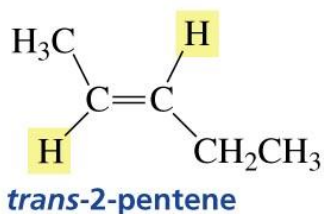


cis-1,4-dimethylcyclohexane

© 2011 Pearson Education, Inc.



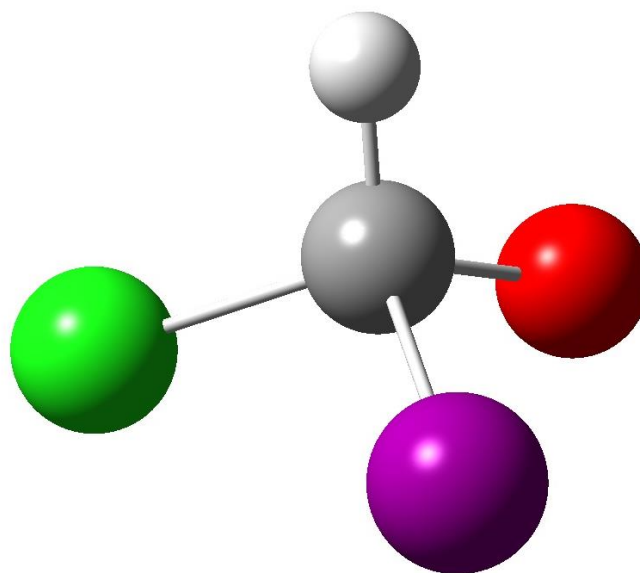
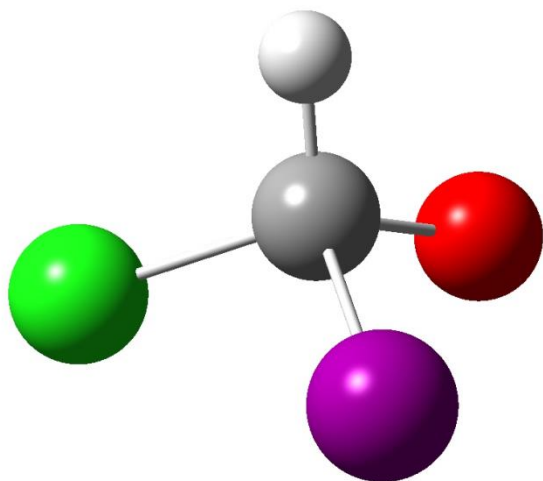
trans-1,4-dimethylcyclohexane

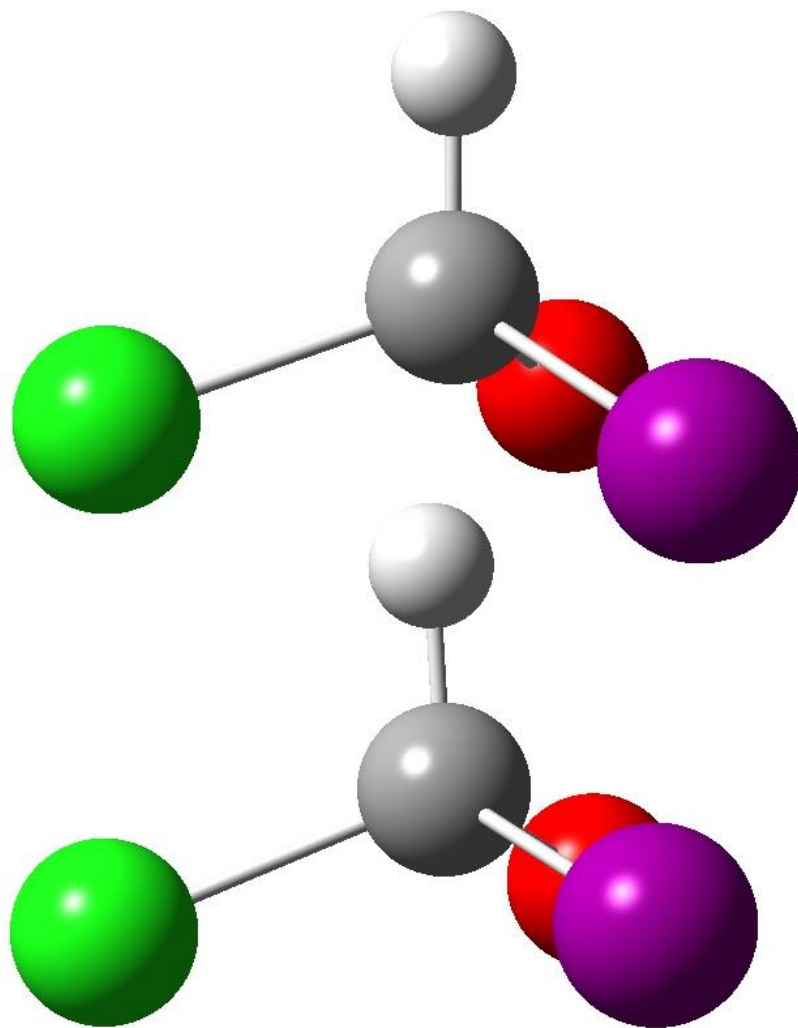


© 2011 Pearson Education, Inc.

© 2011 Pearson Education, Inc.

Are they identical molecules?



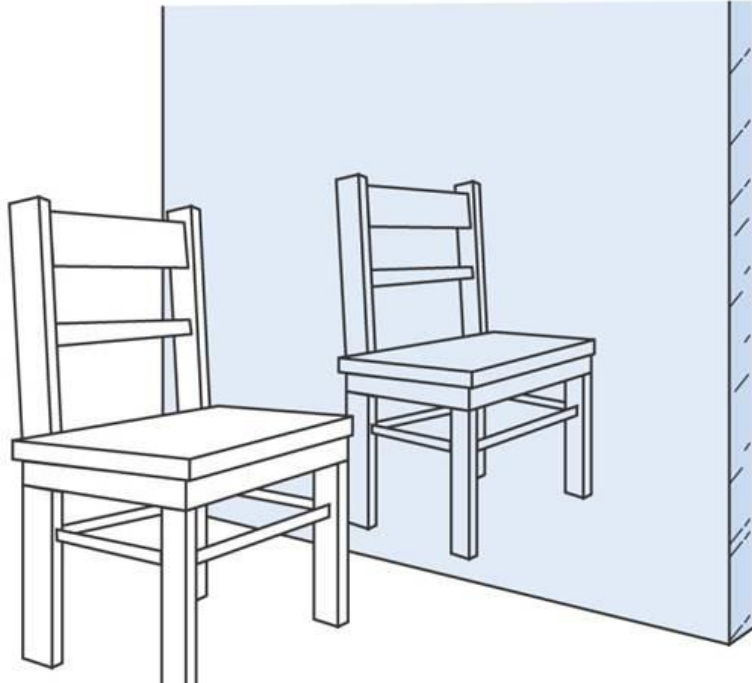


**Superimposable
molecules**

They are identical

“Handedness”: right glove doesn’t fit the left hand.

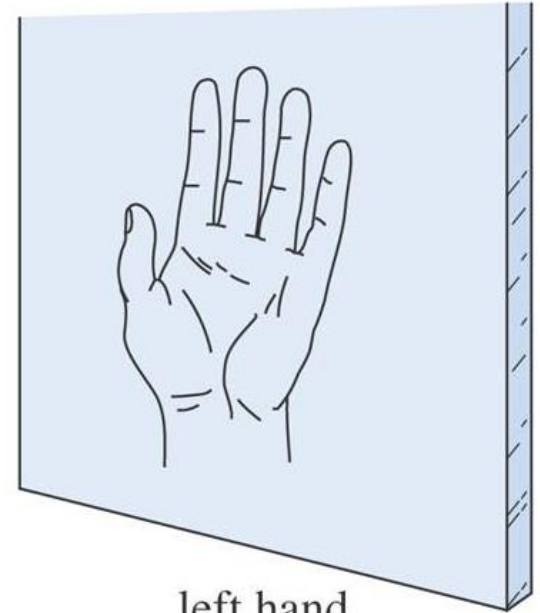
Mirror-image object is different from the original object.



Is the chair identical with its mirror image?



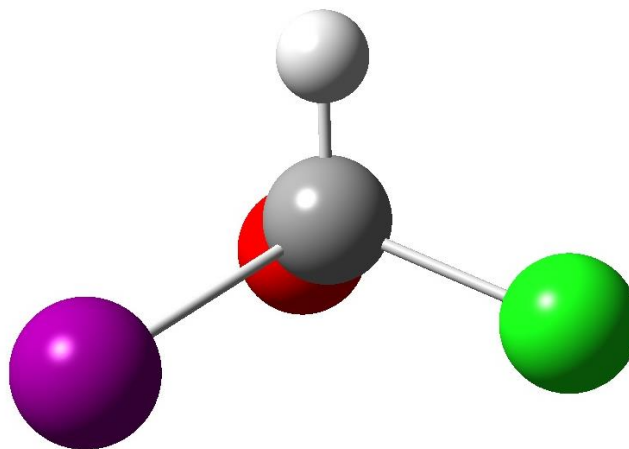
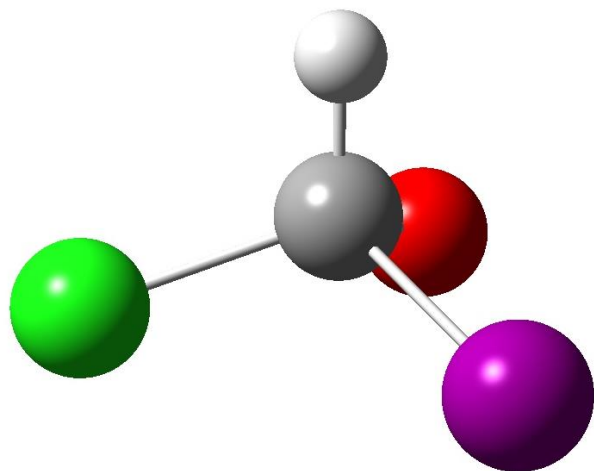
right hand



left hand

Is the left hand identical with right hand?

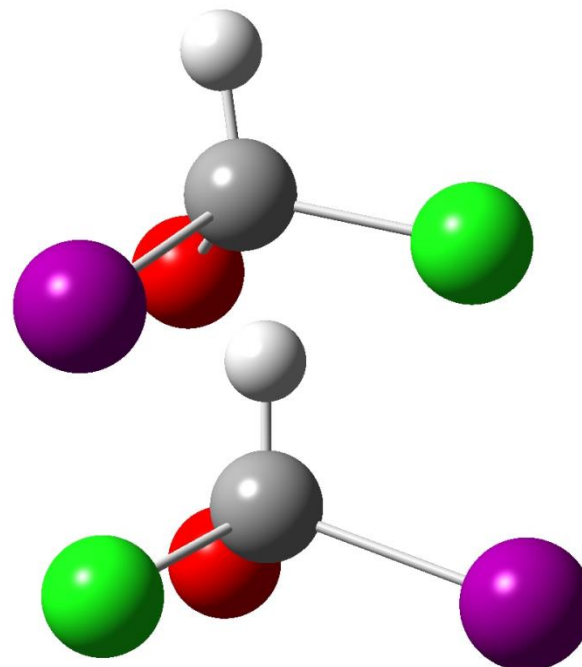
mirror



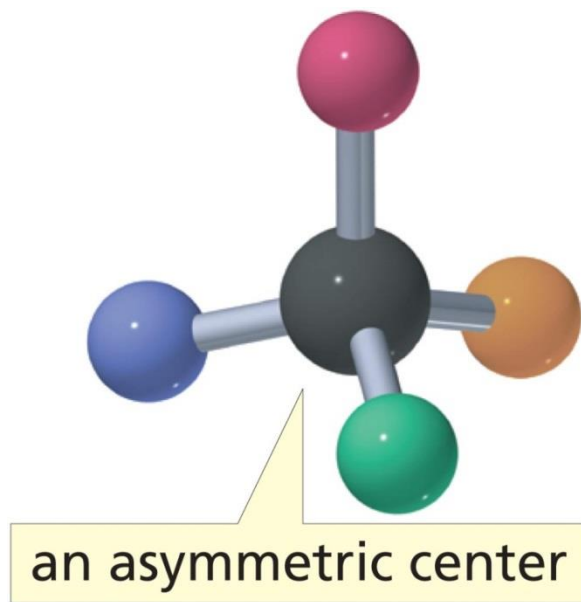
Mirror image

nonsuperimposable

mirror image



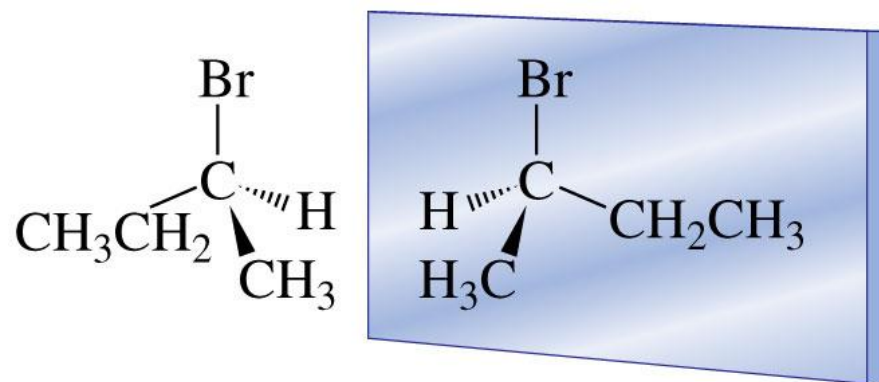
An Asymmetric Center Is a Cause of **Chirality**



Asymmetric or stereogenic center is sp^3 carbon with 4 different substituents

Achiral compounds have superimposable mirror images.

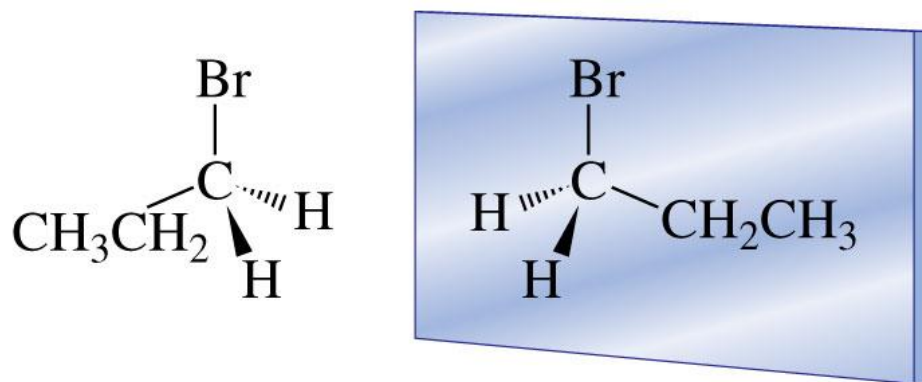
Chiral compounds have nonsuperimposable mirror images.



a chiral
molecule

nonsuperimposable
mirror image

enantiomers

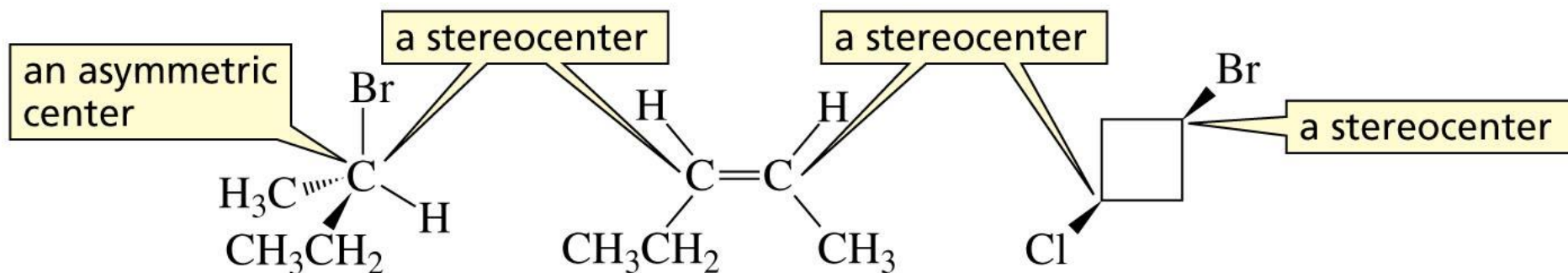


an achiral
molecule

superimposable
mirror image

identical molecules

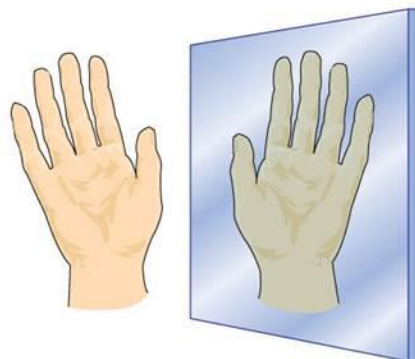
A stereocenter (or stereogenic center) is an atom at which the interchange of two groups produces a stereoisomer



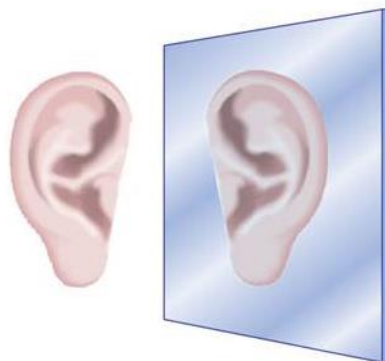
© 2011 Pearson Education, Inc.

Common Objects Also Exhibit **Chirality** or **Achirality**

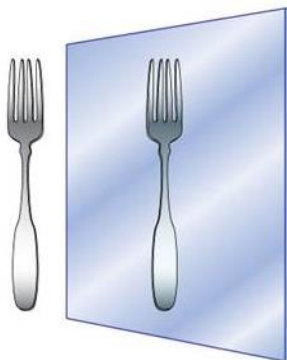
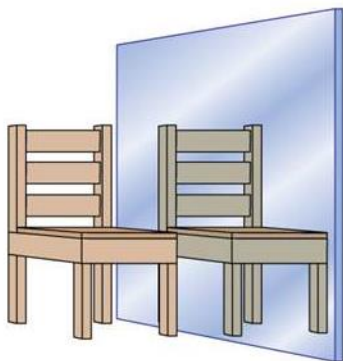
chiral object



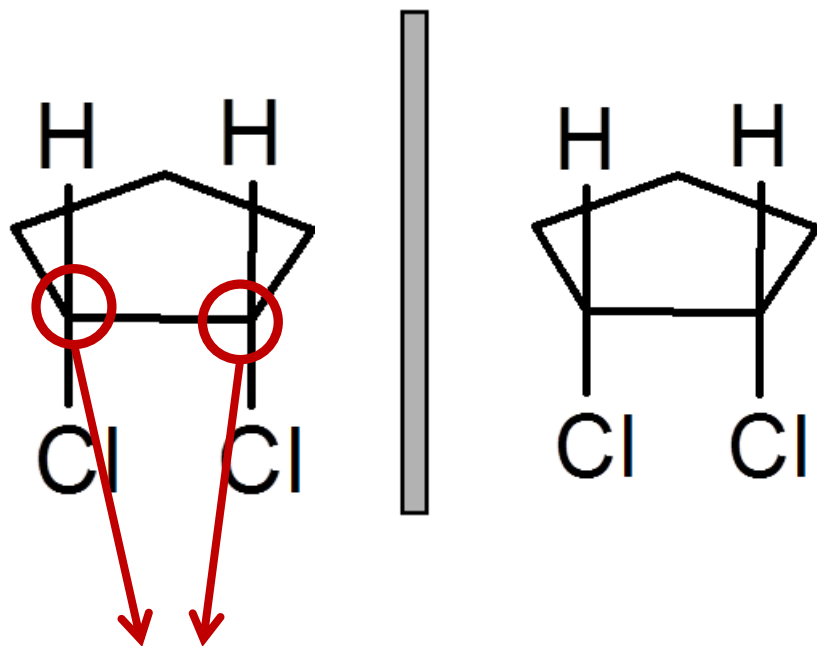
right hand left hand



achiral object



Chiral Molecules

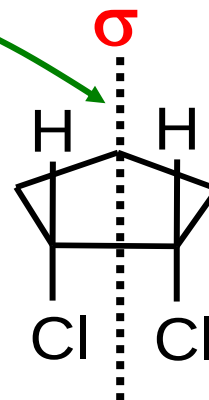


Asymmetric carbon

Cis-1,2-dichlorocyclopentane contains two asymmetric carbons but is achiral.

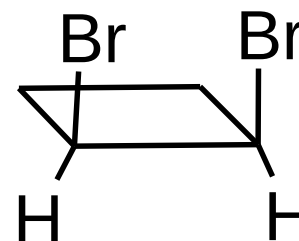
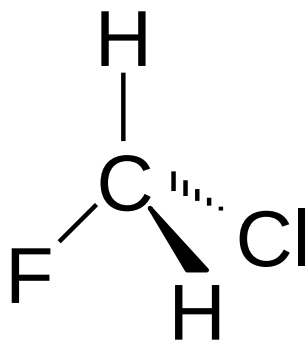
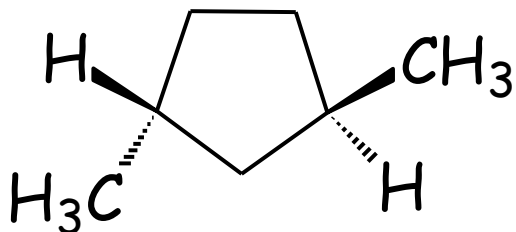
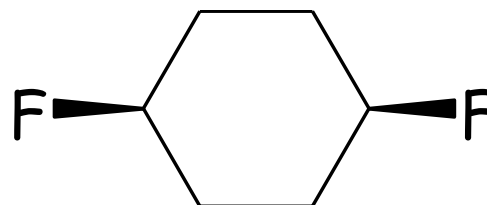
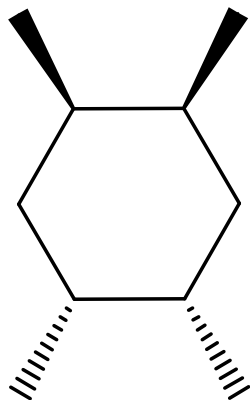
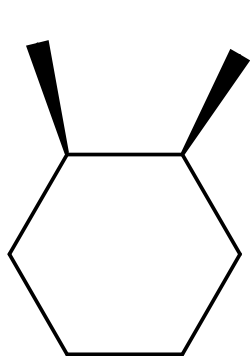
Internal Plane of Symmetry

internal mirror plane of symmetry

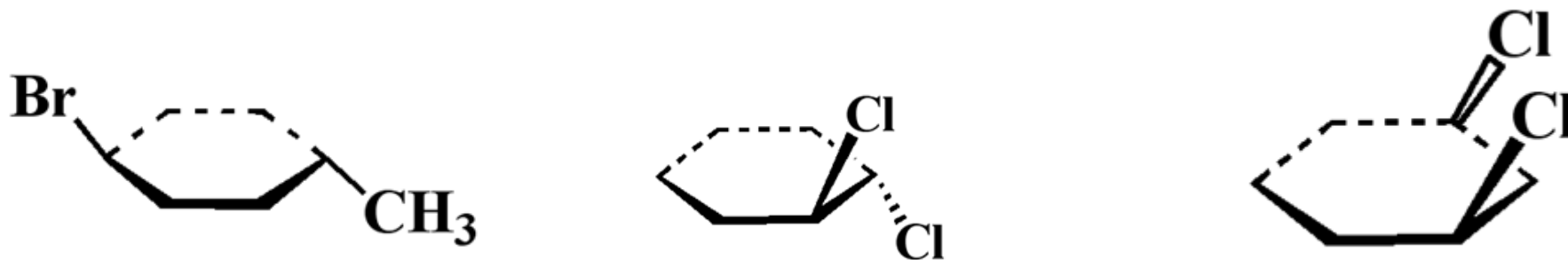


Any molecule that has an internal mirror plane of symmetry is achiral even if it contains asymmetric carbon atoms.

Internal Plane of Symmetry

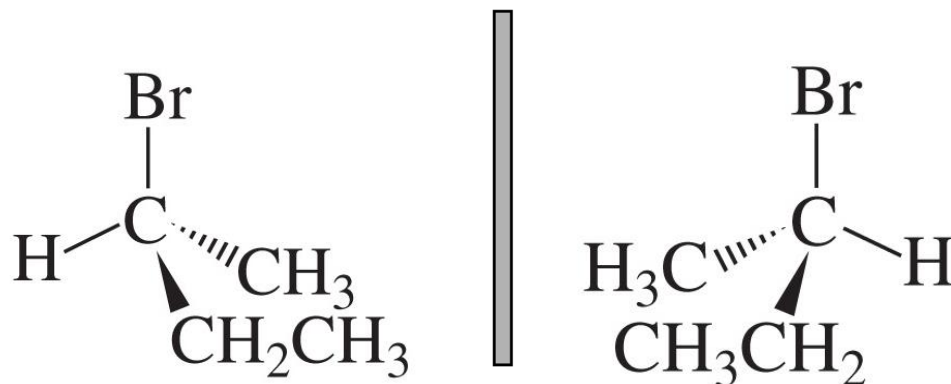


Internal Plane of Symmetry

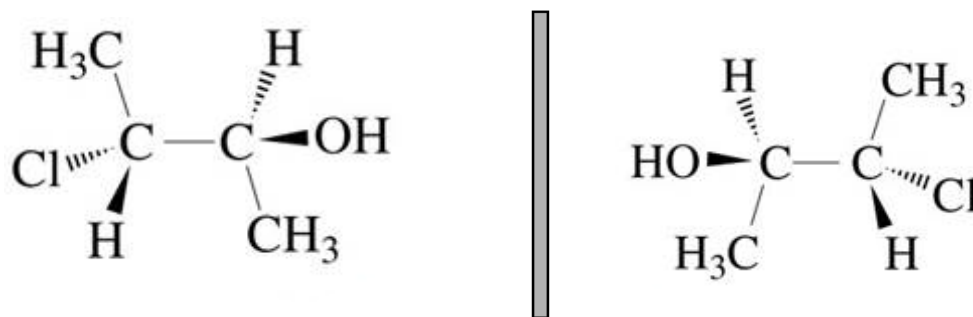


The presence of asymmetric centers does not imply chirality!

Perspective formula



perspective formulas of the enantiomers of 2-bromobutane

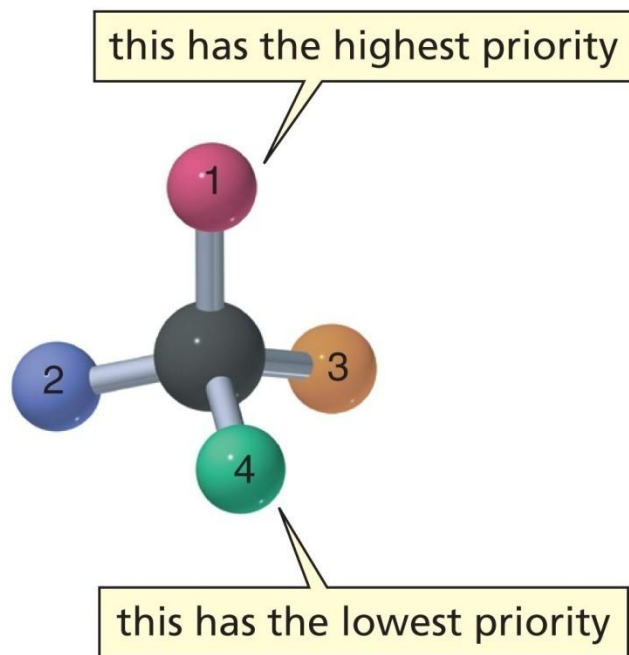


Enantiomers

Naming Enantiomers

The *R,S* system of nomenclature

Rank the groups (atoms) bonded to the asymmetric center.

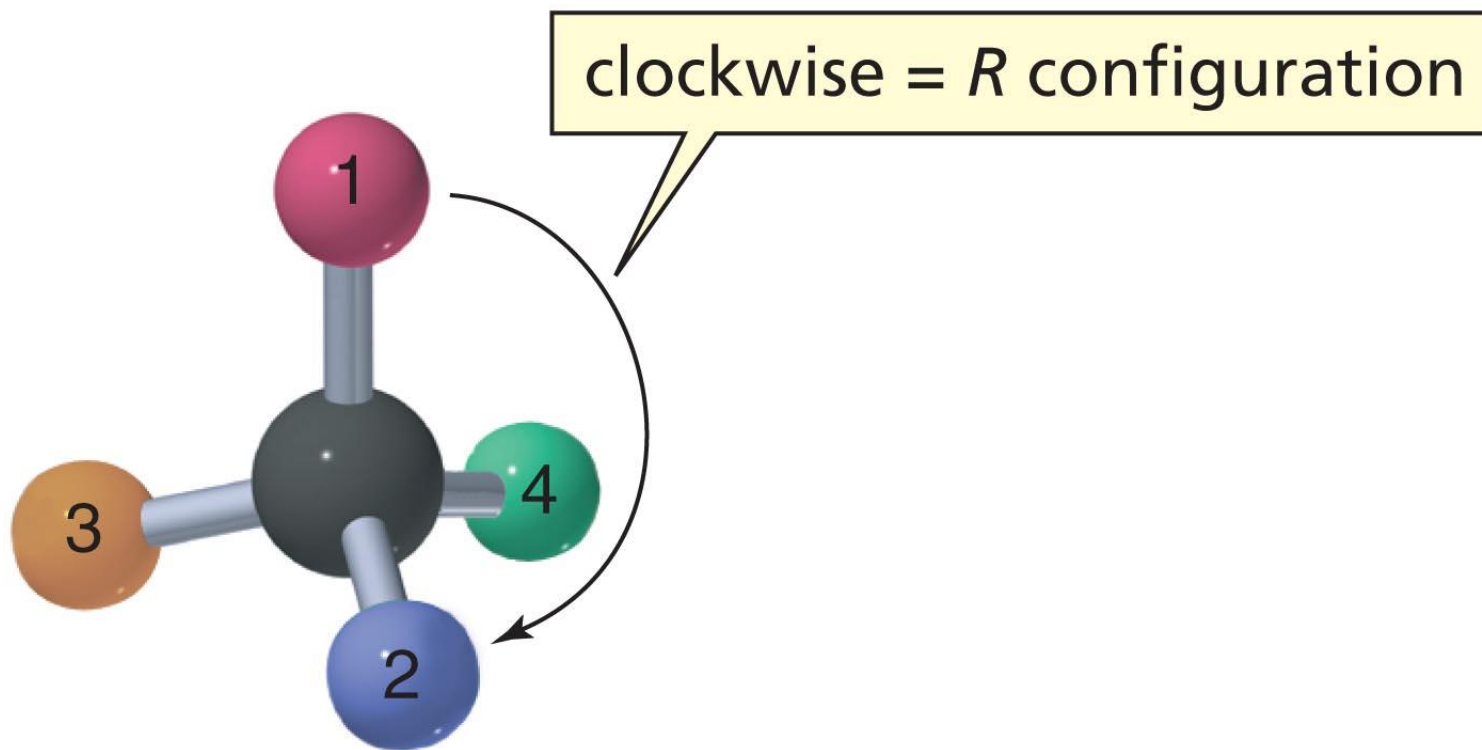


Ranking Rules:

1. Consider the atomic number of the atoms bonded directly to the asymmetric carbon.
2. If there is a tie, consider the atoms attached to the tied atoms.
3. Multiple bonds are treated as attachment of multiple single bonds using "divide-duplicate."
4. Rank the priorities by mass number in isotopes.

Same rules as for *E/Z* assignments

Orient the lowest priority (4) away from you:

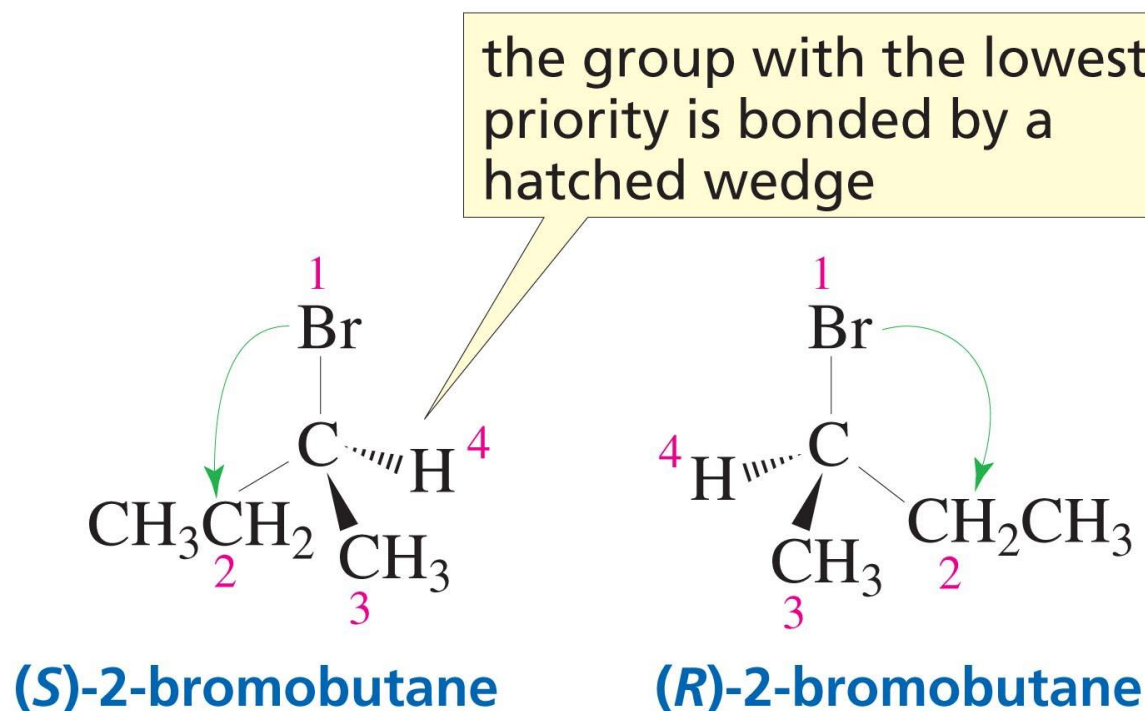


© 2011 Pearson Education, Inc.

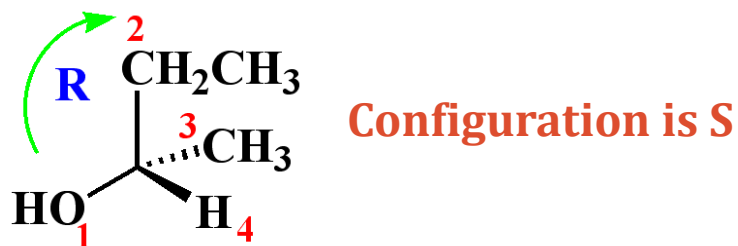
- **Clockwise = *R* configuration**
- **Counterclockwise = *S* configuration**

Naming from the Perspective Formula

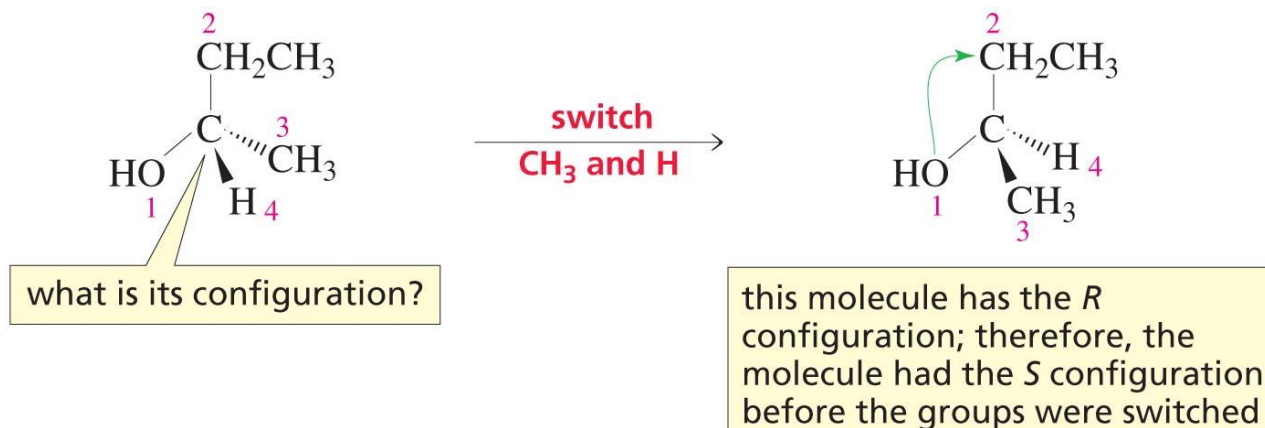
1. Rank the groups bonded to the asymmetric center with the lowest priority group in the back.



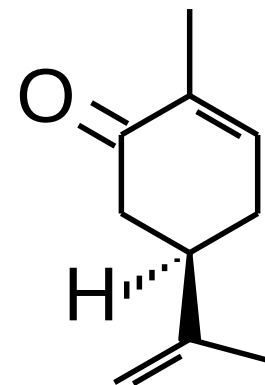
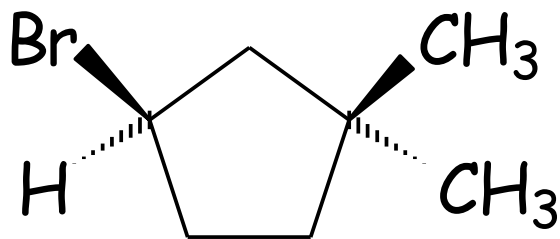
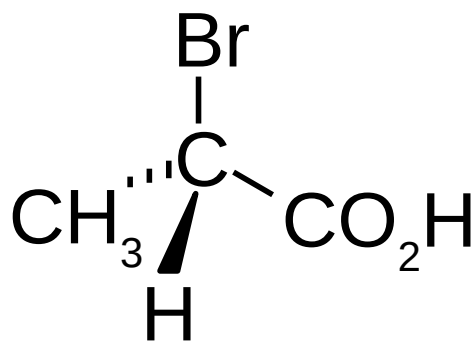
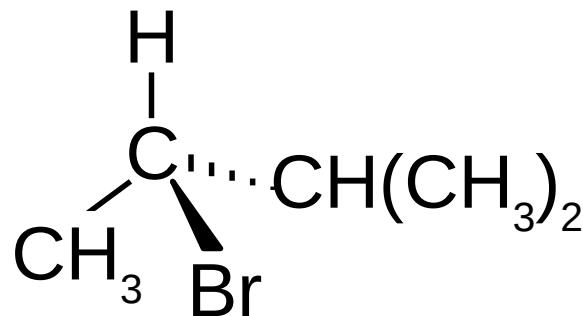
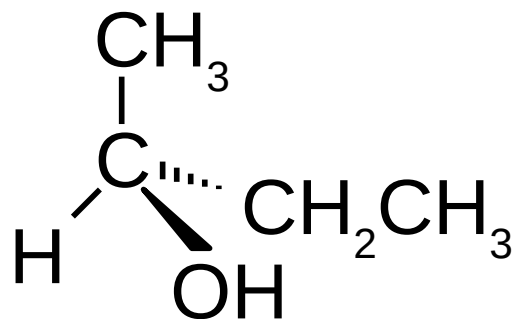
2. If the group (or atom) with the lowest priority is in the front, assign S or R and then switch your answer to R or S respectively.



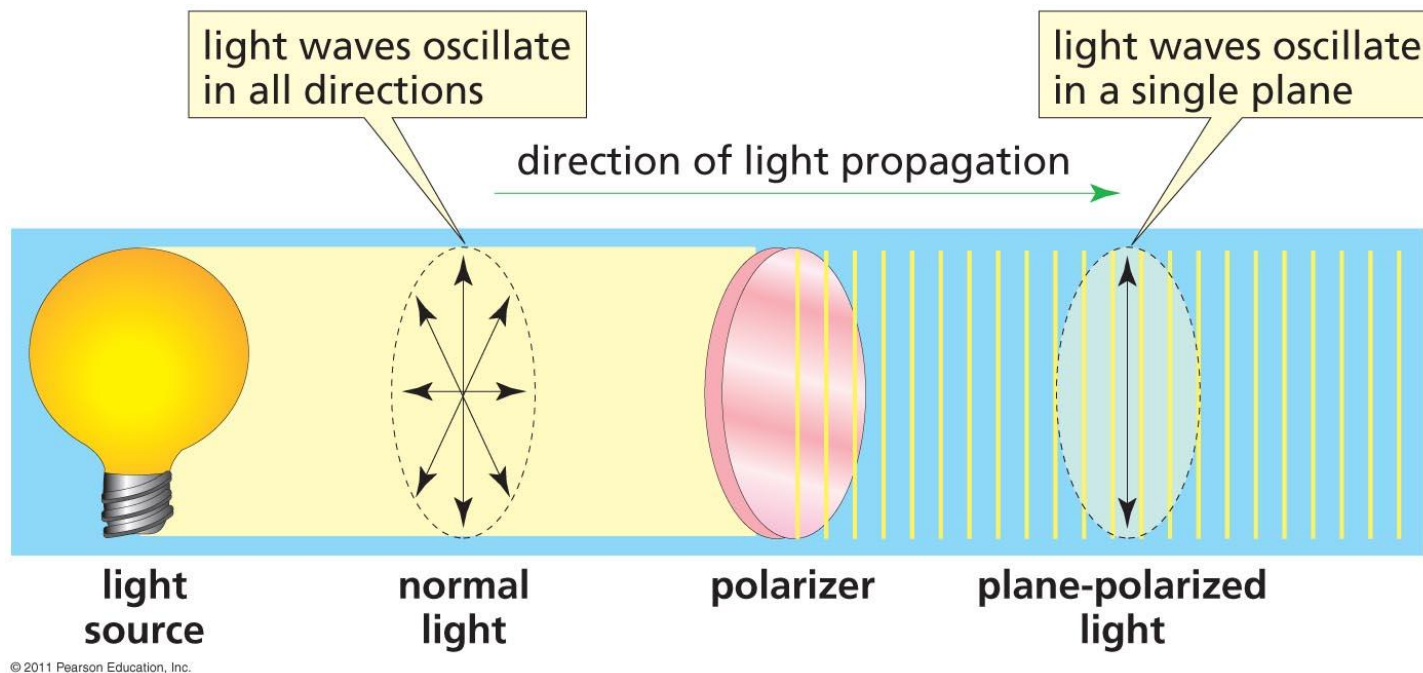
3. Alternatively, atoms or groups can be switched so as to place the lowest priority group in the back. One switch: configuration opposite; two switches, configuration unchanged.



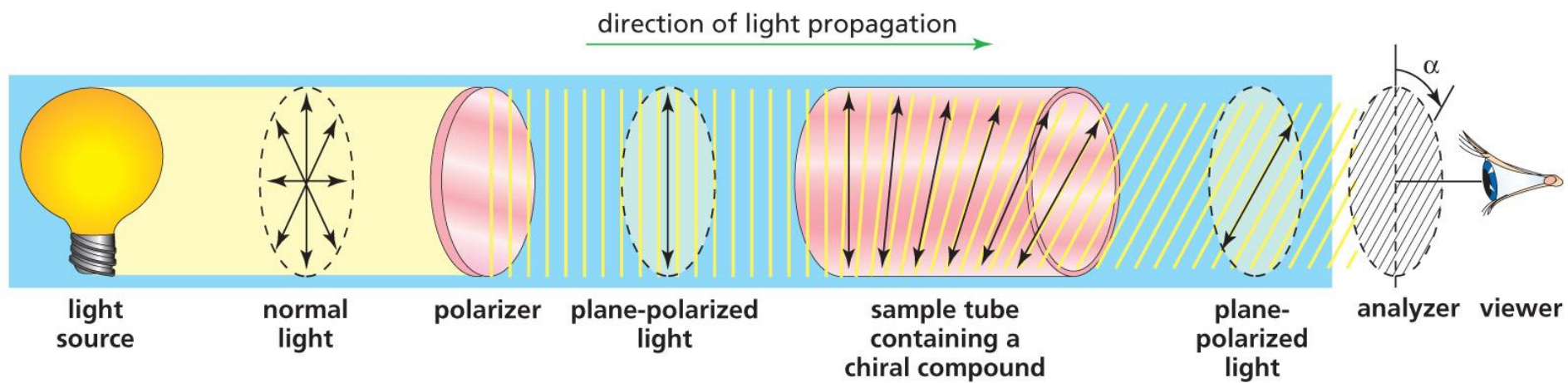
Determine in the following compounds whether it has the (R) or (S) configuration.



Chiral compounds are optically active; they rotate the plane of polarized light.



Achiral compounds do not rotate the plane of polarized light. They are optically inactive.



Copyright © 2006 Pearson Education, Inc.

A polarizer measures the degree of optical rotation of a compound

The observed rotation (a):

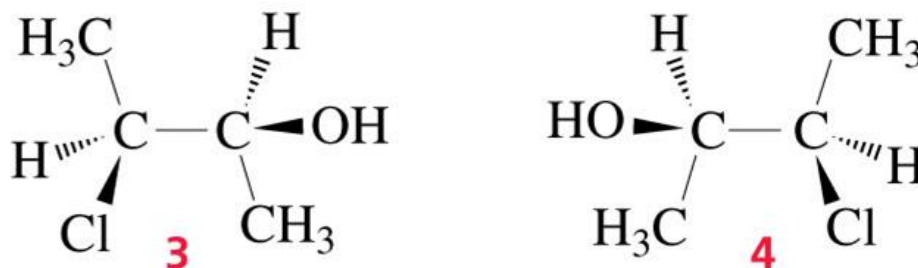
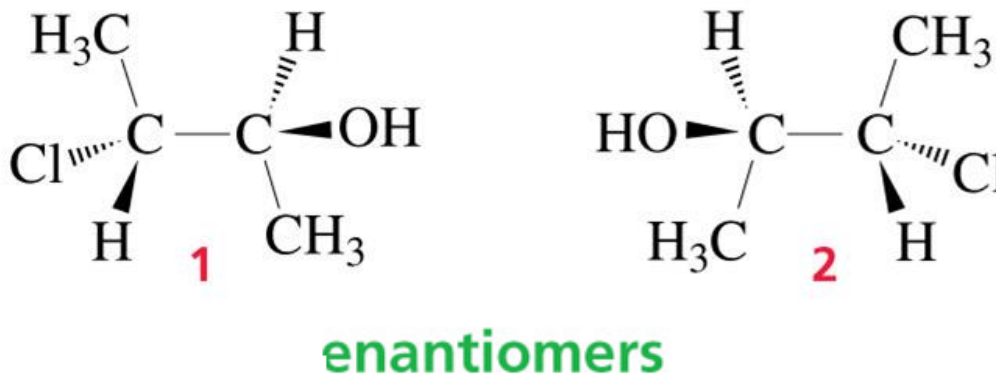
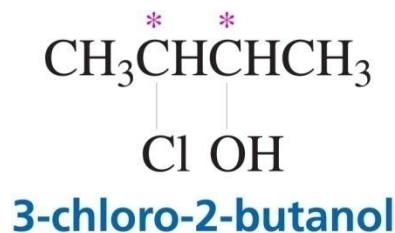
$$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$$

$[\alpha]_{\lambda}^T$ = Specific Rotation

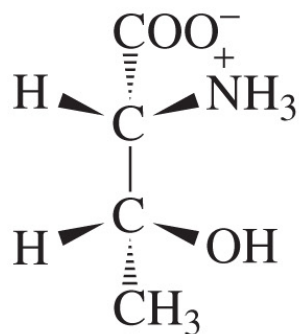
- T is the temp in °C
- λ is the wavelength
- α is the measured rotation in degrees
- l is the path length in decimeters
- c is the concentration in grams per mL

Each optically active compound has a characteristic specific rotation.

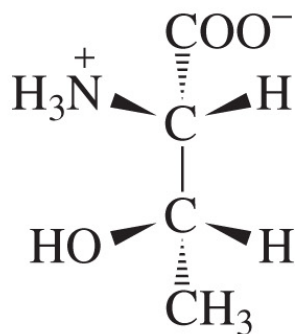
Isomers with more than one asymmetric center: a maximum of 2^n stereoisomers can be obtained



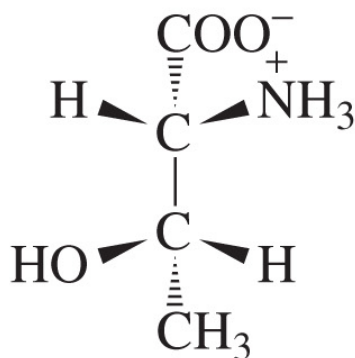
Diastereomers are stereoisomers that are not **enantiomers**



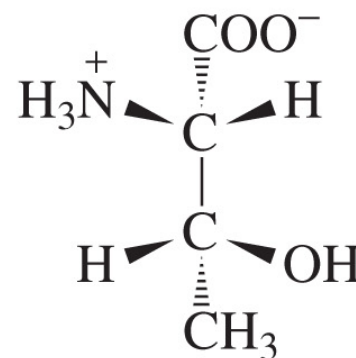
1



2



3



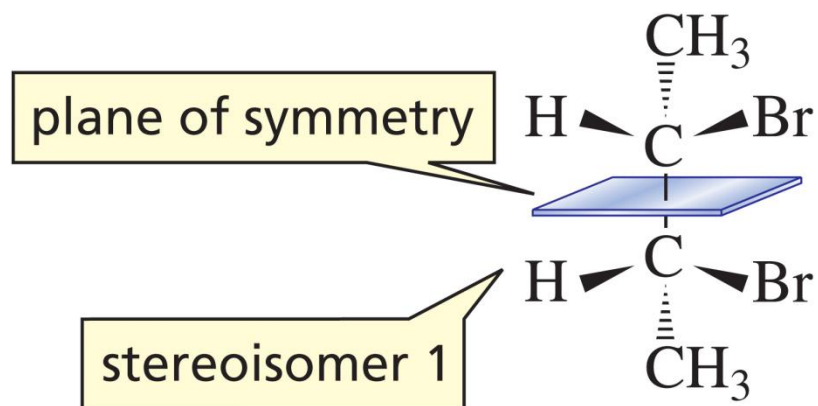
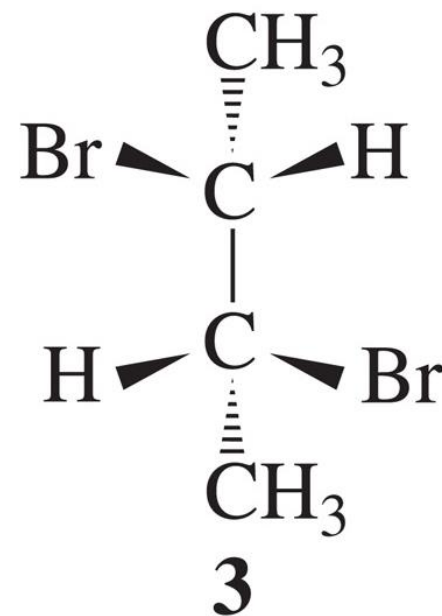
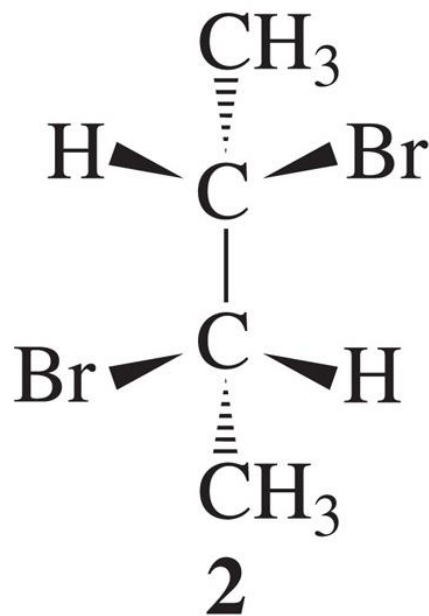
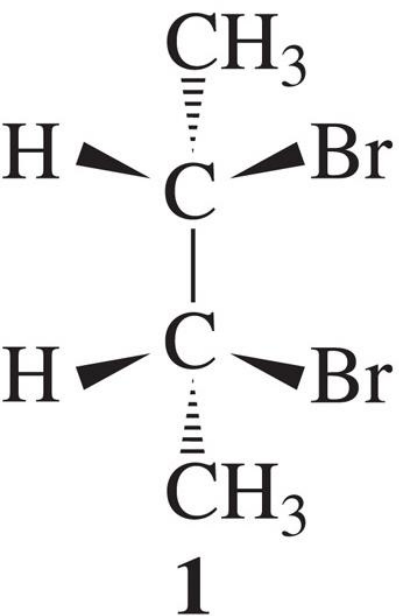
4

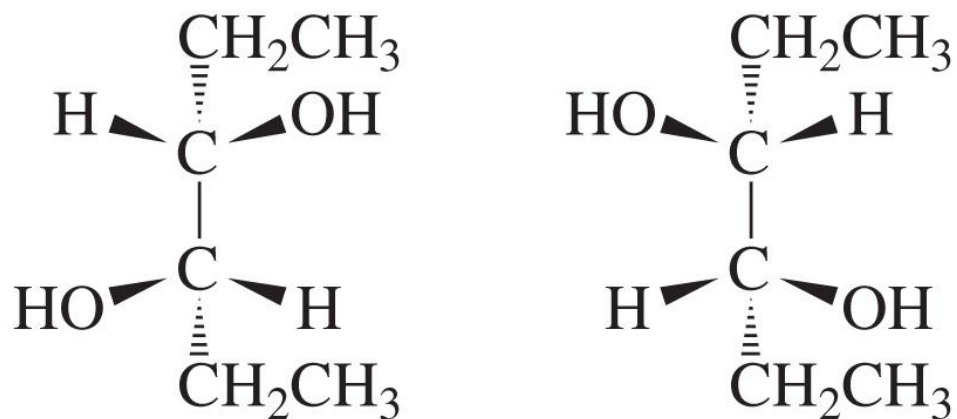
enantiomers

enantiomers

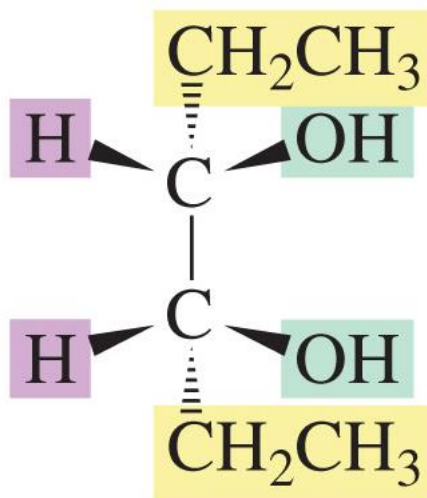
2 and 3 are not **enantiomers**

They are **diastereomers**





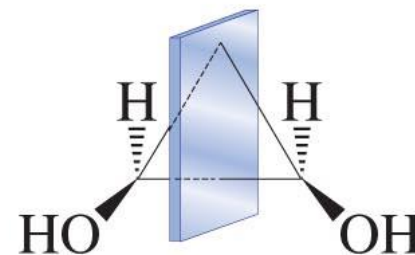
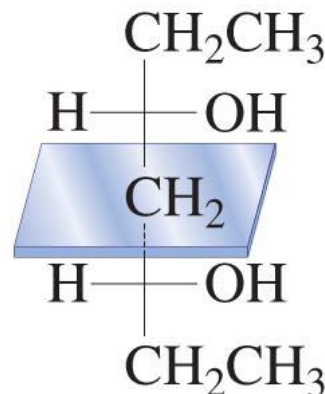
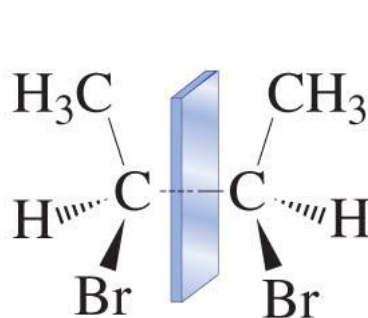
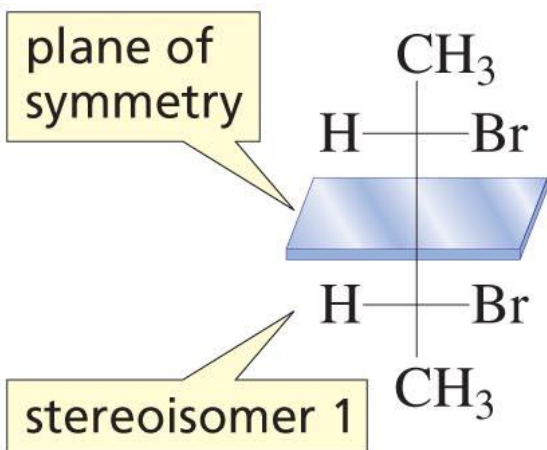
enantiomers



meso compound

Meso Compounds

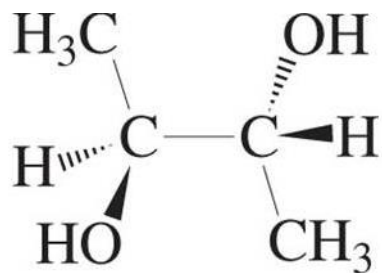
Have two or more asymmetric centers and an internal plane of symmetry (σ -plane)



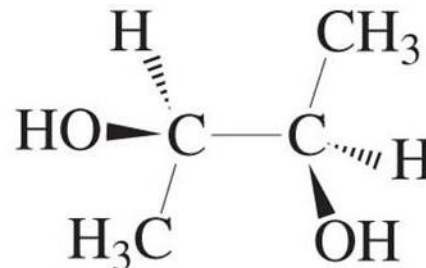
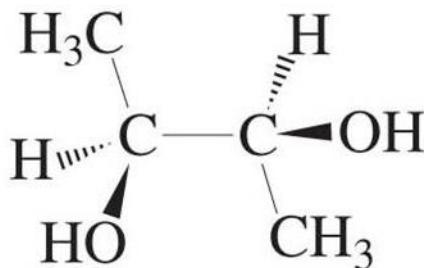
meso compounds

Meso compounds are achiral molecules with asymmetric centers

If a compound with two asymmetric centers has the same four groups bonded to each of the centers, one of its stereoisomers will be a meso compound

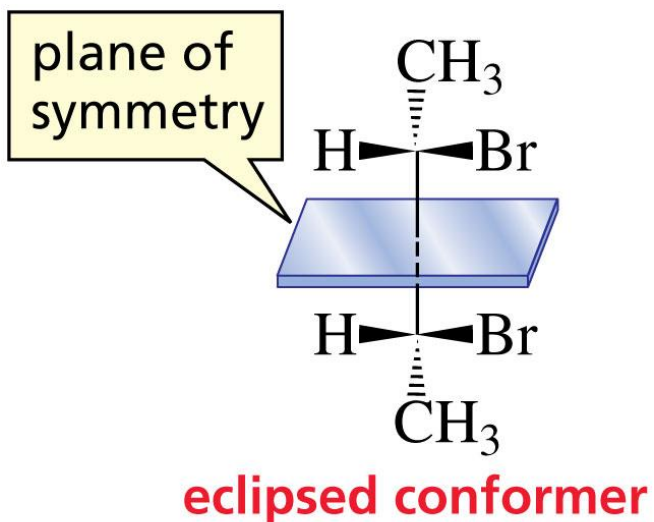


meso compound

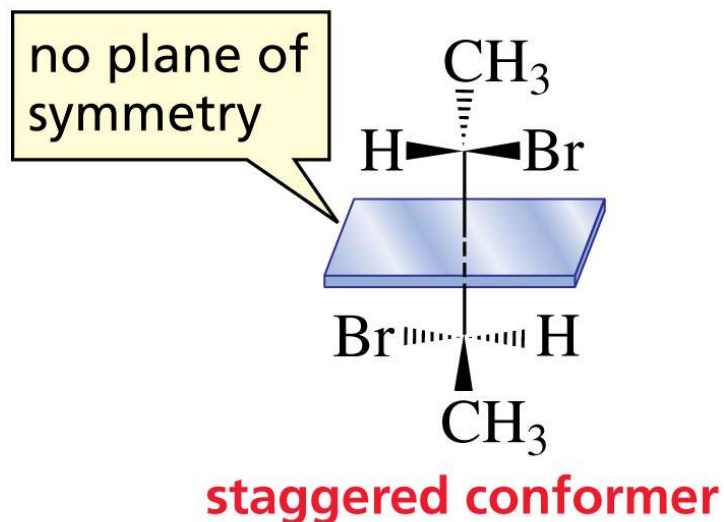


enantiomers

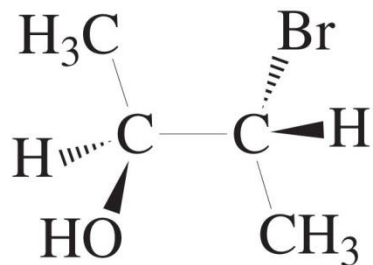
As long as any one conformer of a compound has a plane of symmetry, the compound will be achiral



© 2011 Pearson Education, Inc.



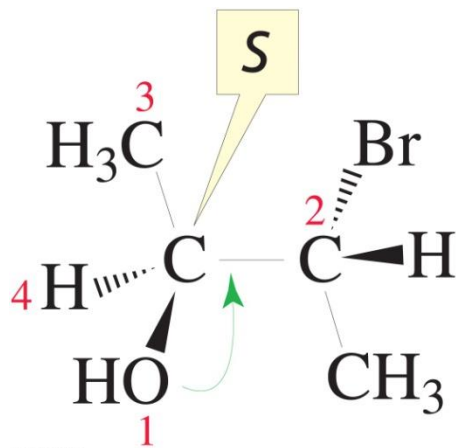
Naming Isomers with More Than One Asymmetric Center



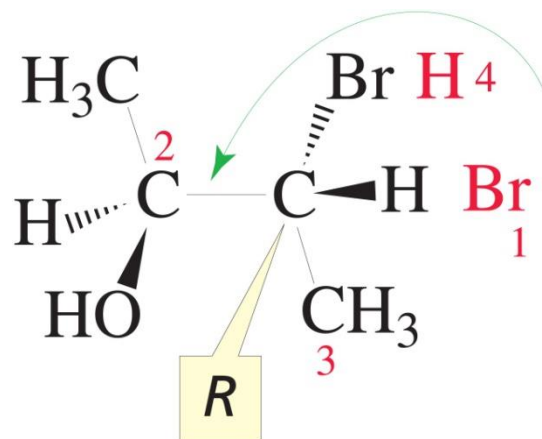
a stereoisomer of 3-bromo-2-butanol

© 2011 Pearson Education, Inc.

The OH group at C-2 has the highest priority, followed by Br in C-3



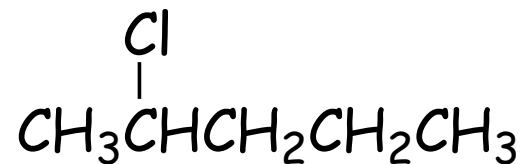
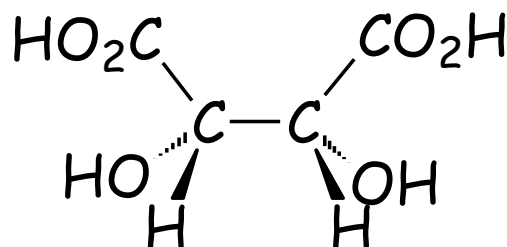
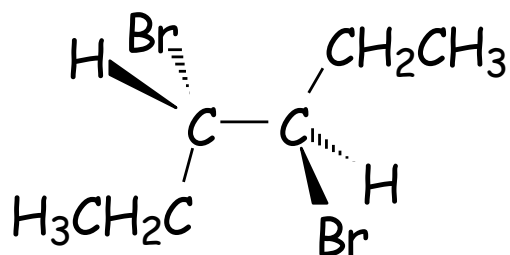
© 2011 Pearson Education, Inc.



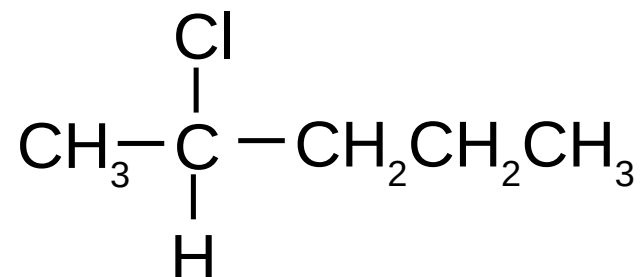
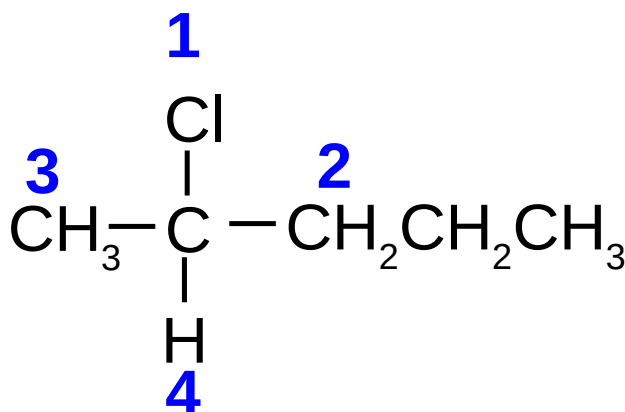
© 2011 Pearson Education, Inc.

The isomer is named (2S, 3R)-3-bromo-2-butanol

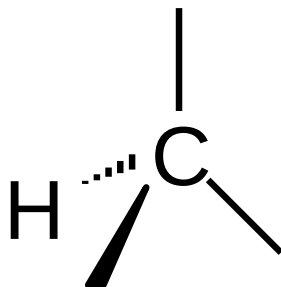
Which of the following compounds are chiral?



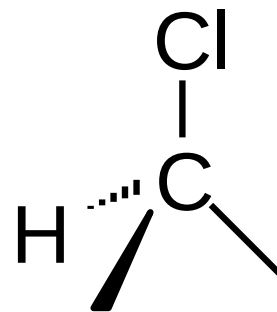
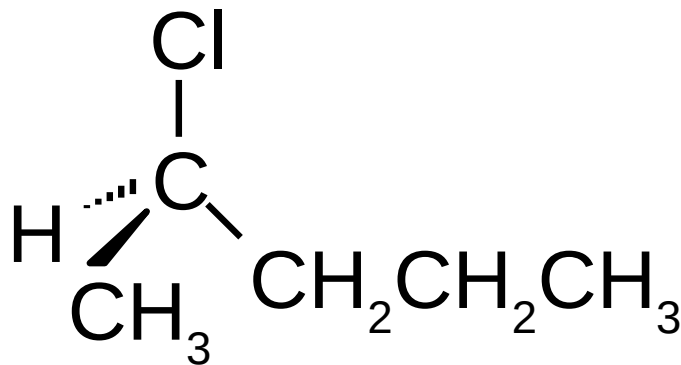
Example: Draw a 3-dimensional formula for (R)-2-chloropentane.



Step 3: Draw a "skeleton" with the asymmetric carbon in the center and the lowest priority group attached to the "dashed" wedge (i.e. pointing away from you).

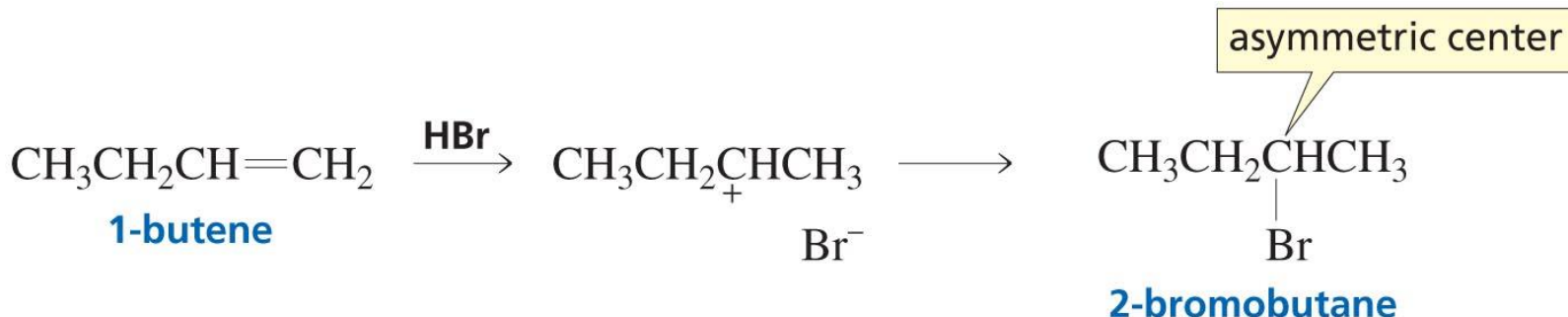


Step 4: Place the highest priority group at the top.

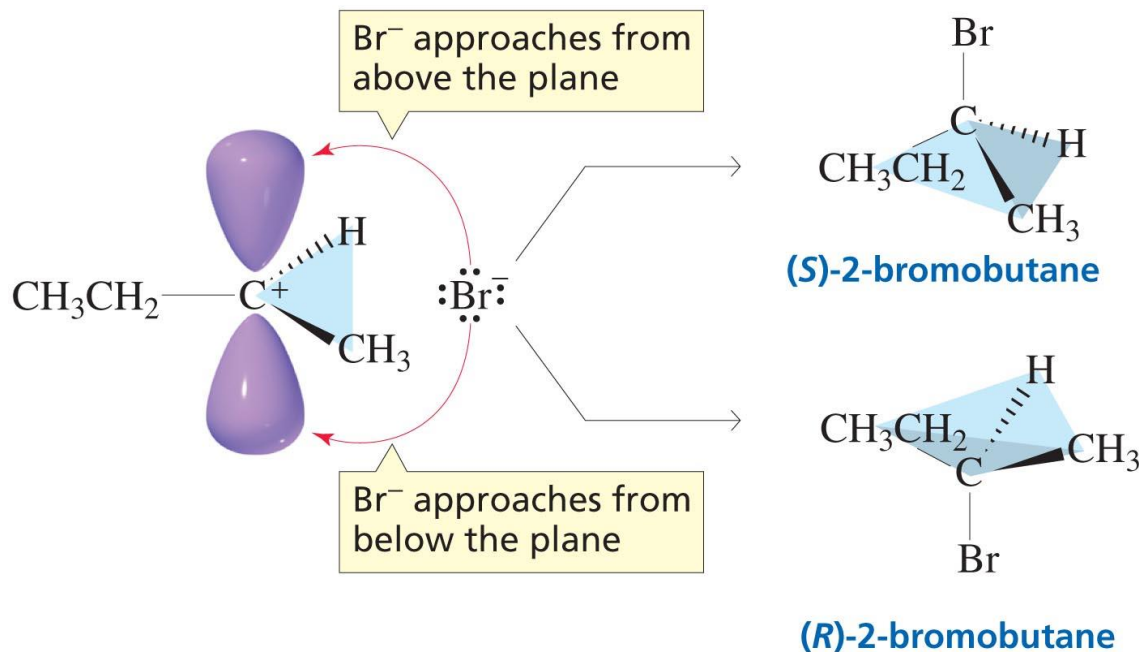


Stereochemistry of Electrophilic Addition Reactions of Alkenes

A regioselective reaction without stereoselectivity



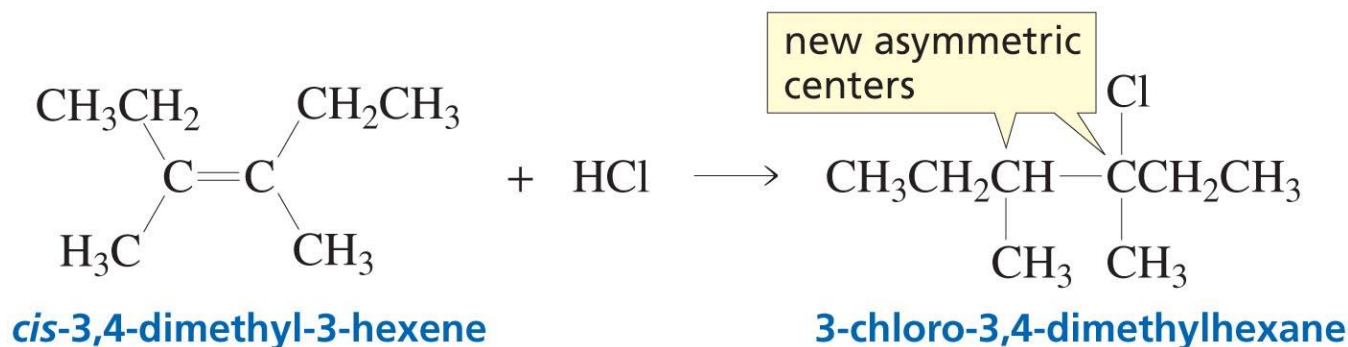
© 2011 Pearson Education, Inc.



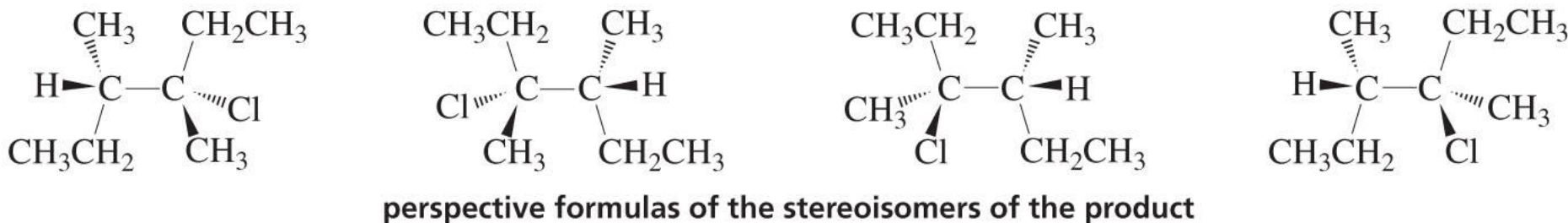
Racemic Mixture Results

© 2011 Pearson Education, Inc.

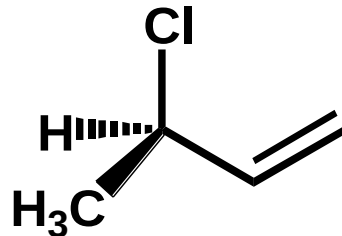
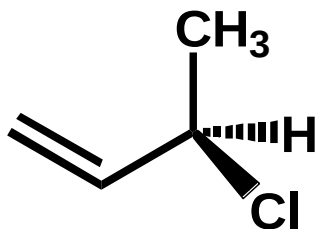
Addition reactions that form products with two asymmetric centers



All four possible stereoisomers result:

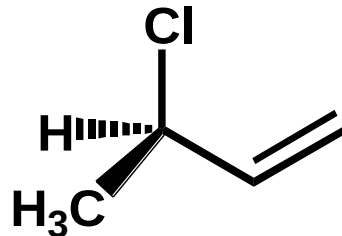
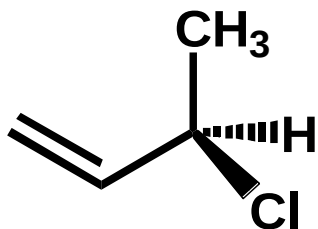


Q: What is the relationship between the two compounds shown below?



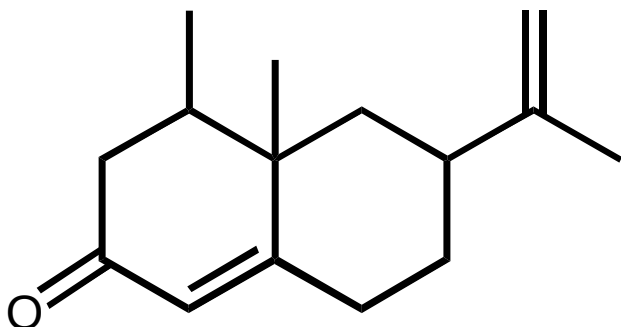
- A. Enantiomers
- B. Diastereomers
- C. Constitutional isomers
- D. Geometric isomers
- E. Identical compounds

Q: What is the relationship between the two compounds shown below?



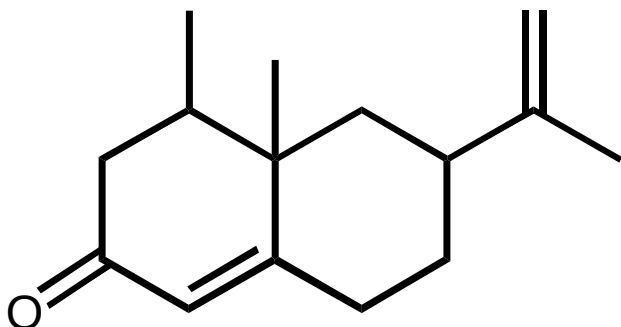
- A. Enantiomers
- B. Diastereomers
- C. Constitutional isomers
- D. Geometric isomers
- E. Identical compounds**

Q: What is the maximum number of stereoisomers possible for nootkatone (grapefruit oil) shown below?



- A. 2**
- B. 3**
- C. 4**
- D. 8**
- E. 16**

Q: What is the maximum number of stereoisomers possible for nootkatone (grapefruit oil) shown below?



- A. 2
- B. 3
- C. 4
- D. 8**
- E. 16